

Module 3 Task: Operation Overcast - Phase II

Role: Lead Cloud Forensic Engineer

Objective: Deconstruct a highly obfuscated cloud compromise involving data exfiltration and credential theft.

1. Scenario

Your organization's AWS environment has been hit by a sophisticated actor. Unlike previous attacks, this actor used obfuscation to hide their findings in the logs. You have been provided with the CloudTrail events.

The Challenge

The logs contain a mix of standard developer activity, automated AWS system checks, and malicious actions. Some critical information has been encoded to evade basic pattern-matching tools.

2. Requirements & Deliverables

Your investigation report must provide a complete timeline of the breach. Submit a standard and structured report. Your report should contain images and steps you followed.

To be successful, you must answer the following:

1. **Credential Discovery:** Identify the Access Key ID that was stolen and used for the bulk of the attack.
2. **IP Forensic Tracking:** What was the Initial Attacker IP?
 - What was the Second IP used for persistence?
 - Provide the timestamp of the first action taken from the second IP.
3. **Obfuscated Data Analysis (Signal vs. Noise):**
 - There are two distinct Base64 encoded strings in the logs. Decode both.
 - Which one is legitimate and which is not?
 - What can you deduce from the decoded strings?
4. **Administrative Escalation:** Identify the specific timestamp and API call where the attacker granted themselves full administrative privileges.
5. **The Theft:** Which specific file was downloaded from S3? Provide the Bucket name and the File name.
6. **Persistence & Evasion:** Name the new Backdoor User created by the attacker.

- What was the final action the attacker took against the original stolen data? Why would an attacker do this?

3. Tooling Requirements

You can use:

- **CLI Parsing:** jq (Linux/WSL) or PowerShell (ConvertFrom-Json).
- **Decoders:** CyberChef or the base64 command line tool.
- **Graphical Interface:** JSONCrack.